

STEVENS POINT, WI, USA

WMO: 726426

Lat: 44.550N

Lon: 89.533W

Elev: 1110

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-7.8	-19.8	1.9	-11.0	-15.2	2.5	-6.2	19.2	22.7	17.6	22.7	3.1	280	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	88.4	72.4	84.7	70.0	82.1	68.5	75.8	83.3	73.7	80.9	71.8	79.2	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	128.6	79.4	71.9	123.0	78.2	69.8	114.1	76.2	40.0	84.4	38.0	80.0	36.2	78.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	17.0	15.3	DB	-20.2	92.8	4.1	4.0	-23.2	95.7	-25.6	98.0	-27.9	100.3	-30.9	103.2
			WB	-20.6	78.0	4.1	2.5	-23.5	79.8	-25.9	81.3	-28.2	82.7	-31.1	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	15.9	18.2	30.3	43.6	55.6	64.8	69.7	67.2	59.6	46.6	34.1	20.8
	DBStd	21.23	12.51	11.94	12.04	9.91	8.74	6.79	6.07	6.04	8.29	9.00	10.15	11.49
	HDD50	4434	1058	891	620	241	41	1	0	0	14	178	484	906
	HDD65	8096	1523	1310	1075	645	312	85	24	45	203	575	928	1371
	CDD50	2243	0	0	11	48	216	445	610	533	301	72	7	0
	CDD65	430	0	0	1	2	22	79	169	113	40	4	0	0
	CDH74	4242	0	0	5	46	277	817	1674	1005	375	43	0	0
	CDH80	1186	0	0	0	9	71	218	538	257	88	5	0	0
(14) Wind	WSAvg	6.2	6.1	6.3	6.5	7.8	7.2	6.2	5.4	4.9	5.5	6.2	6.2	6.1
(15) Precipitation	PrecAvg	32.2	1.1	1.0	1.7	3.0	3.9	4.7	3.8	4.0	3.5	2.5	1.8	1.3
	PrecMax	41.0	2.5	2.0	3.4	5.9	7.0	8.5	9.6	11.9	9.2	6.1	5.3	2.9
	PrecMin	26.5	0.4	0.3	0.3	0.8	1.9	1.7	1.0	0.7	0.4	0.2	0.0	0.2
	PrecStd	3.4	0.6	0.4	0.9	1.2	1.2	2.2	2.0	2.2	2.0	1.3	1.2	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	50.4	71.9	80.8	87.8	90.4	94.1	90.5	87.9	79.4	65.8	51.6
		MCWB	39.4	44.2	60.3	62.1	70.2	73.1	74.7	73.7	71.3	64.4	55.6	50.6
	2%	DB	39.2	43.5	61.3	72.5	81.3	85.4	89.8	86.1	82.1	72.7	59.1	43.1
		MCWB	35.8	38.8	51.2	56.8	64.6	69.6	73.4	71.2	67.9	62.6	51.9	39.1
	5%	DB	35.9	38.2	54.3	66.2	76.0	81.9	85.7	82.2	78.9	66.7	54.4	38.5
		MCWB	34.1	35.0	47.7	53.2	60.8	68.0	71.5	69.0	66.3	58.3	47.8	36.4
	10%	DB	33.3	34.5	47.6	61.4	72.0	79.1	82.2	79.7	74.6	62.7	49.2	36.0
		MCWB	32.1	31.9	41.7	49.3	59.4	66.1	69.3	67.5	64.5	55.4	44.8	34.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	45.4	60.0	64.5	72.1	76.6	79.5	78.2	74.2	70.1	58.4	51.2
		MCDB	44.3	49.8	69.8	75.5	82.7	84.3	88.5	84.6	80.7	76.6	63.6	51.5
	2%	WB	36.5	39.3	54.2	59.3	67.7	73.5	76.8	74.6	71.5	64.0	53.4	40.1
		MCDB	39.4	42.8	58.2	68.8	76.6	82.0	84.9	81.3	78.9	68.7	57.7	41.6
	5%	WB	34.0	35.6	47.4	55.1	64.5	71.0	74.5	72.4	69.2	60.2	48.5	36.8
		MCDB	35.5	38.3	54.0	64.5	71.8	79.2	81.7	79.2	75.2	66.2	53.3	38.4
	10%	WB	31.6	32.1	41.6	50.6	61.6	68.4	72.2	70.2	66.6	55.8	44.8	34.1
		MCDB	32.8	34.3	47.6	59.5	68.8	75.6	79.3	76.7	72.3	61.8	48.4	35.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.7	18.8	20.8	21.9	20.9	21.6	20.7	21.5	18.8	15.8	14.2
		MCDBR	16.9	18.7	25.8	30.8	28.0	24.7	24.5	23.1	25.3	25.7	23.9	16.4
	5% WB	MCWBR	14.0	14.2	16.7	17.6	13.4	11.9	10.6	10.8	12.5	15.5	16.9	13.8
		MCDWR	15.4	17.5	23.3	26.3	22.6	21.1	20.9	19.4	20.6	21.9	21.6	14.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.286	0.319	0.352	0.388	0.390	0.382	0.383	0.355	0.319	0.288	0.260
		taud	2.403	2.292	2.370	2.362	2.303	2.326	2.361	2.354	2.431	2.524	2.524	2.449
	Ebn at Noon		282	286	291	289	280	279	280	275	275	273	266	267
		Edh at Noon	23	31	33	37	40	40	38	37	31	25	21	20
(44) All-Sky Solar Radiation	RadAvg		536	887	1246	1441	1721	1884	1929	1649	1308	826	524	391
	RadStd		59	77	133	146	132	138	99	108	97	89	57	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	-1.89	-4.52	-5.22	N/A	N/A	N/A	N/A	N/A	N/A	-144
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page